1 Overview

The HX-200 coolant pump circulates glycol coolant through the inverter cabinets of the Nordhafen solar park and through the backup chiller of the Ostfeld dialysis centre. Each pump is rated for 42 litres per minute at a differential pressure of 2.4 bar. The pump is driven by a 1.1 kW motor with an integrated frequency converter.

This manual is valid for serial numbers HX2-1000 to HX2-1999. Older units with the grey terminal box use the procedures in the legacy manual HX-150.

2 Safety

Isolate the pump electrically before opening the terminal box. Lock out breaker Q12 and attach a personal lock. Wait five minutes for the DC link capacitors of the converter to discharge.

Coolant can reach 65 degrees Celsius. Wear heat resistant gloves and eye protection when venting the system.

3 Fault codes

The converter shows a three digit fault code on its display. Codes starting with E1 relate to the hydraulic side, codes starting with E2 to the electrical side.

Fault E104 means dry running: the pressure at the inlet fell below 0.3 bar for more than ten seconds. The pump stops and restarts automatically after 120 seconds for at most three attempts.

Persistent faults require an inspection of the maintenance history before the pump is restarted.

Code	Meaning	First action
E104	Dry running	Check strainer S3
E117	Cavitation	Reduce speed to 80 %
E203	Heat sink over-temperature	Clean fins, check fan F2
E219	Phase loss	Measure phases at X1

3.1 Hydraulic faults

If E104 repeats, check the inlet strainer S3 for clogging and verify that the expansion vessel pre-charge is 1.5 bar. A clogged strainer is the most common cause of E104 in the first year of operation.

Fault E117 indicates cavitation detected by the vibration sensor. Reduce the speed set point to 80 percent and inspect the impeller for pitting at the next planned maintenance window.

3.2 Electrical faults

Fault E203 is an over-temperature of the converter heat sink above 85 degrees Celsius. Clean the cooling fins and check that the cabinet fan F2 is running.

Fault E219 signals a phase loss on the supply. Measure all three phases at terminal X1 and tighten the terminals to 1.2 Nm.

4 Maintenance schedule

Every 3 months: inspect the strainer S3, check for leaks at the mechanical seal and read out the fault log of the converter.

Every 12 months: replace the coolant filter cartridge, check the expansion vessel pre-charge and verify the calibration of pressure sensor P1.

Every 5 years or 30000 operating hours: replace the mechanical seal and the bearings of the motor.

5 Contacts

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